

PAPER_1627 — Gas Constant $R = R = K_MEX \cdot (D - F^2) = (25/12) \cdot 3.99 = 8.3125 \text{ J}/(\text{mol} \cdot \text{K})$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Thermodynamics **Date:** June 18, 2026 **Status:** CLOSED — 0.018% closure

Observation

PAPER_1209AA_S584 (Thermodynamics Unified Proof Set) gives:

$$R = K_MEX \cdot (D - F^2) = (25/12) \cdot 3.99 = 8.3125 \text{ J}/(\text{mol} \cdot \text{K})$$

Residual: **0.018%**.

UQFF Closed Identity

$$R = K_MEX \cdot (D - F^2) = (25/12) \cdot 3.99 = 8.3125 \text{ J}/(\text{mol} \cdot \text{K}) \quad 0.018\%$$

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical thermodynamics measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209AA_S584
- Related: PAPER_1494-1625 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "gas_constant_r_8_314"})`

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